HEALTHIER, LONGER, BETTER LIVES

Medical Receipt Extraction

Benchmark results and pilot direction

OCR rules vs Azure OpenAI · 9 receipts · 6 fields

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Executive Summary

Recommendation: Azure DI + AOAI prompt v2 for the next receipt extraction pilot, not production.

96.1%

Azure DI + AOAI (prompt v2)

Development result

51.0–70.6%

OCR + multilingual rules

51.0% PaddleOCR Light · 70.6% Azure DI

Retrospective baseline

What we recommend

- Azure DI + AOAI prompt v2 for the next extraction pilot
- Python validation: format checks today, missing-field checks planned
- Only approved receipt data on Azure; audit checks stay in another authorized system
- **A pilot, not a production approval**

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Scope and Data Boundary

PLANNED WORKFLOW · NOT BUILT

Azure only reads receipts and extracts data. Audit checks happen in another authorized system.

Approved Azure environment: receipt data only

Medical receipt
approved receipt only

→

Azure DI
reads the text

→

Azure OpenAI
extracts the fields

→

Python checks
existing: formats, review flags

→

Structured results
e.g. names, date, amounts

Today: Azure DI + Azure OpenAI on 6 fields, with existing Python checks (date and amount formats, label clean-up, review flags).
Planned: required-field checks and checks for the new fields.

Approved transfer
to be designed

→

Another authorized system

- Holds claims records, policy data and claim history
- Runs the audit checks: claim comparison, policy checks, duplicates

Never uploaded to Azure: claims records, policyholder databases, policy documents or other sensitive business information.

Planned workflow. The transfer method must be approved before implementation; the integration and the audit checks are not being built yet.

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Benchmark: Three Approaches, Same 9 Receipts

MEASURED · DEVELOPMENT SET

Multilingual OCR rules reached 51.0–70.6%; Azure DI + AOAI reached 96.1%.

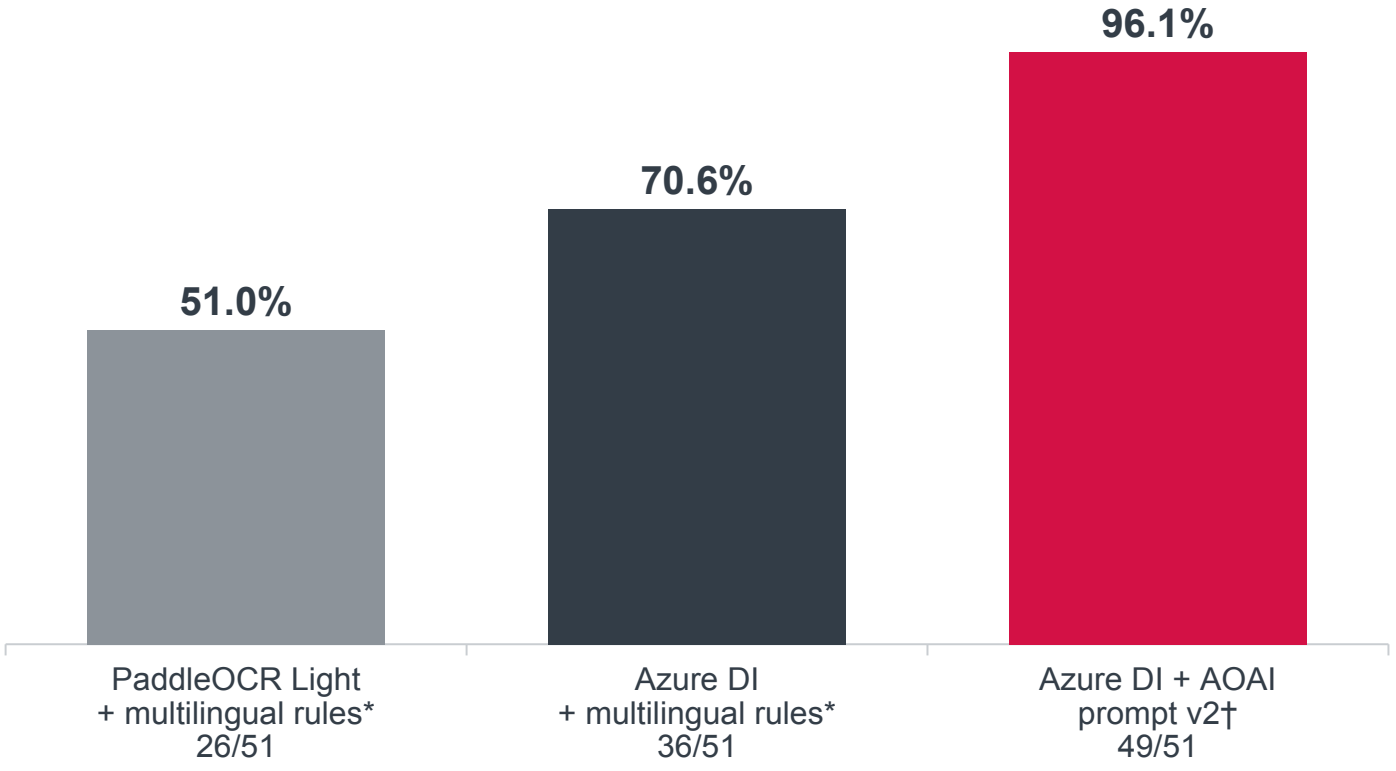
PaddleOCR Light + rules
Open-source OCR → Hand-written rules

Azure DI + rules
Azure OCR → Hand-written rules

Azure DI + AOAI
Azure OCR → Azure OpenAI

9 receipts · English, bilingual, Chinese layouts

6 fields: doctor, registration no., patient, receipt no., date, total



** Retrospective baseline † Development result Neither is independent validation. 51 of 54 fields scored (3 receipts have no registration number).*

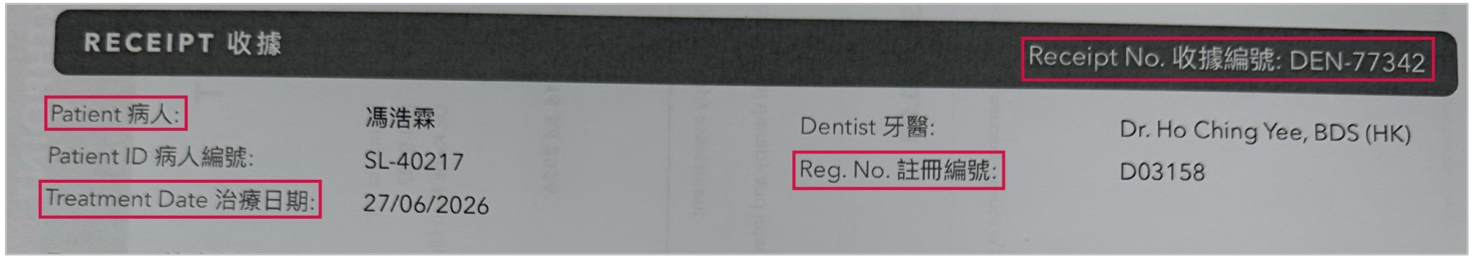
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Why Azure DI Beat PaddleOCR Light

MEASURED · RETROSPECTIVE

Same rules on both. PaddleOCR misread the labels, so the rules could not find the values.


RECEIPT 收據

Receipt No. 收據編號: DEN-77342

Patient 病人: 馮浩霖

Patient ID 病人編號: SL-40217

Treatment Date 治療日期: 27/06/2026

Dentist 牙醫: Dr. Ho Ching Yee, BDS (HK)

Reg. No. 註冊編號: D03158

10 extra errors with PaddleOCR

All from reading the label text: words run together, wrong characters, a misread HK$ sign. None from the rules or the layout.

On this receipt PaddleOCR lost 4 fields: patient, registration no., treatment date and receipt no.

Field	Correct value	PaddleOCR read the label as…	Azure DI read it as…
Registration no.	D03158	✗ Reg.No. 註冊編號 *words run together → not found*	✓ Reg. No. 註冊編號 → *found*
Treatment date	27/06/2026	✗ TreatmentDate 治療日期 *words run together → not found*	✓ Treatment Date 治療日期 → *found*
Receipt no.	DEN-77342	✗ Receipt No. 收據编號 *编 read for 編 → not found*	✓ Receipt No. 收據編號 → *found*

Same multilingual rules (retrospective baseline) on both engines. Receipt 5 is a bilingual dental receipt.

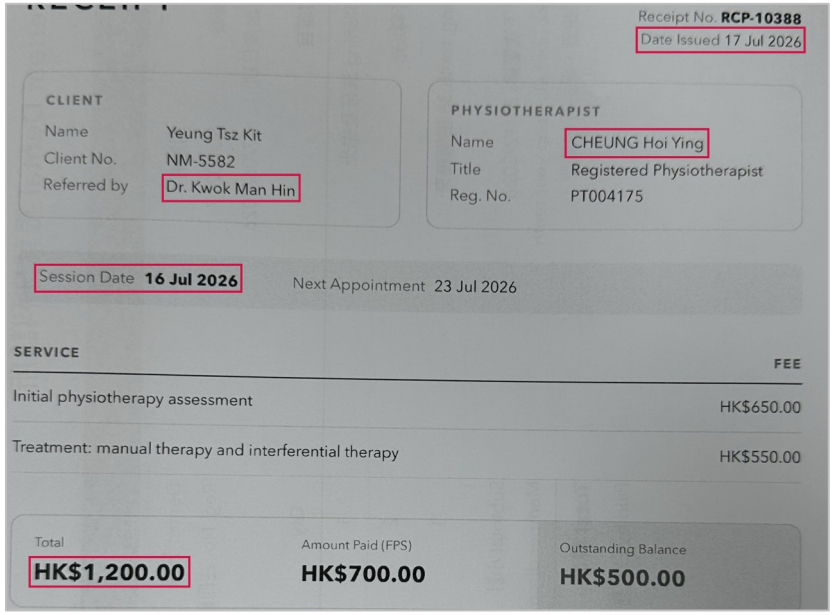
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Why AOAI Beat the Rules

MEASURED · RETROSPECTIVE

Both read the same Azure DI text. AOAI used the context; the rules followed fixed patterns.



Field	Correct	Rules	AOAI
Practitioner Receipt 6	CHEUNG Hoi Ying	✗ Dr. Kwok Man Hin *took the referring doctor*	✓ CHEUNG Hoi Ying
Service date Receipt 6	16 Jul 2026	✗ 17 Jul 2026 *took the date the receipt was issued*	✓ 16 Jul 2026
Total Receipt 6	HK$1,200	✗ not found *could not read the card layout*	✓ HK$1,200
Total Receipt 5	HK$1,575	✗ HK$1,750 *added up the items, ignored the discount*	✓ HK$1,575

Rules: logic written case by case

Unfamiliar labels, layouts or scenarios may need additional rules and retesting (tuning these 9 grew the rules from 69 to 225 label phrases).

AOAI: plain-language instructions

Can often handle varied wording without new code (not guaranteed); one prompt revision fixed 4 fields and broke none.

AOAI was right with its original prompt (written before these receipts) and with prompt v2. Rules = Azure DI + multilingual rules (retrospective baseline).

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The Doctor Database Approach

PROPOSAL · NOT BENCHMARKED

Exact matching can identify registered doctor names in OCR text, but cannot determine who treated the patient.

The proposal

1. **OCR reads the receipt**
   Azure DI or PaddleOCR
2. **Match against the registry**
   ~40,000 doctors · ~80,000 English and Chinese name variants
3. **Strict exact match**
   No fuzzy matching, no AI

Straightforward; no experiment needed
Low cost · Deterministic · Easy to verify

Only works if the OCR reads the name exactly and the registry contains that practitioner.

But a name match is not the treating doctor

Seen in our receipts	
Referring doctor	Receipt 6: “Referred by Dr. Kwok Man Hin”; the treatment was by a physiotherapist (also receipt 2)
Several doctors or staff	Receipt 8: a doctor and a nurse. Vet invoice: a clinician and a signing vet
OCR misreads the name	Signature “Dr. Lok Yi Choi” read as “Dr. Lok Vichol”: no exact match

Hypothetical (not in our receipts)	
Patient is a registered doctor	*The patient's own name matches the registry*
Clinic director in the header	*A director's name printed in the letterhead*

AOAI reads context such as “Referred by” (slide 6) but has not been tested on these cases. Names on our test receipts are fictional, so registry match rates were not measured.

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Advantages and Disadvantages

Rules win on cost, auditability and predictability; AOAI on accuracy, maintenance and flexibility.

	OCR + rules	Azure DI + AOAI	Evidence
Cost	+ PaddleOCR: own compute, no API fee (Azure DI: page fee)	− Azure DI page fee + ~962 input / 243 output tokens per receipt	*Prices to confirm*
Speed	PaddleOCR 13.9 s (CPU) · Azure DI 3.8 s	6.4 s per receipt	Measured
Accuracy	− 51.0–70.6% with multilingual rules (retrospective)	+ 96.1% with prompt v2 (development result)	Measured
Maintenance	− Unfamiliar labels or layouts may require additional rules and retesting	+ Many changes made in plain-language instructions; still retest	*Measured / expected*
Flexibility	− Finds only what it was written for	+ New fields can be requested in the prompt (not yet benchmarked)	*Expected*
Auditability	+ Every result traceable to a rule	− Harder to explain; keep quoted evidence and logs	*Expected*
Reliability	+ Same input, same output	− Can return a confident wrong value (e.g. a sub total); needs Python validation and retesting	*Measured / expected*

Green = stronger side. Multilingual rules include common Chinese labels (retrospective baseline). Expected = not measured here.

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What the Auditors Check

We asked the business auditors what they check when sampling claims, and what information they need.

1

Receipt authenticity

Letterhead, provider details, signature, stamp, signs of alteration

2

Data accuracy and identity

Names, treatment date, diagnosis, amounts; the right patient

3

Coverage and eligibility

Policy coverage, waiting periods, deadlines, preapproval

4

Necessity and fees

Medically necessary treatment, agreed medical fees

5

Assessment and fraud indicators

Calculations and limits, approval authority, duplicates

Why it matters: these checks tell us which receipt information is worth extracting.

Summarised from our discussion with the business auditors. Not every check uses receipt data.

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From Audit Needs to Receipt Fields

On Azure we extract what the receipt shows. Another authorized system adds the rest and runs the checks.

On the receipt (extracted on Azure)	Supports	Status
Patient name	Accuracy	**Benchmarked**
Doctor / practitioner name	Accuracy	**Benchmarked**
Registration number	Authenticity, Accuracy	**Benchmarked**
Receipt number	Fraud	**Benchmarked**
Treatment date	Coverage	**Benchmarked**
Receipt total	Necessity & fees, Fraud	**Benchmarked**
Clinic name and address	Authenticity	*Not yet tested*
Individual charges	Necessity & fees, Fraud	*Not yet tested*
Diagnosis, symptoms (if stated)	Necessity & fees	*Not yet tested*
Treatments, tests, medicines	Necessity & fees	*Not yet tested*
Signature / stamp present	Authenticity	*Not yet tested*

approved transfer

Another authorized system adds

- Claimant, policy and coverage
- Claimed and preapproval amounts
- Claim history, fee schedules, decisions

…and runs the audit checks

Information, not a verdict

Extraction does not decide whether a receipt is genuine, a treatment necessary, or a claim covered.

Later: analytics (potential)

Cost by treatment, provider comparison, trends, unusual claims

Green = the 6 benchmark fields (96.1%, development result). Other fields are not yet tested and appear only if printed or visible on the receipt.

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Recommendation and Next Steps

Pilot Azure DI + AOAI prompt v2 for receipt extraction, then grow the extracted fields step by step.

Next extraction pilot: Azure DI + AOAI prompt v2, with Python validation

A direction for further testing, not a production approval. Only approved receipt data on Azure.

1. Define the schema

Which receipt fields, formats and “if printed” rules, from the auditors' checklist

2. Extend the prompt

Add fields a few at a time; keep the 6 benchmark fields stable

3. Test new fields

On approved receipt examples, including receipts not used for development

4. Prepare the hand-off

Agree the approved transfer method; confirm costs

Rules still fit simple, clearly labelled checks, e.g. whether a diagnosis comment is present.

Evidence so far: 9 receipts; development and retrospective results, not independently validated. No new fields or audit checks are implemented yet.

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